

Constraint Satisfaction Problems



Defining Constraint Satisfaction Problems

- A constraint satisfaction problem consists of **three components, X , D , and C** :
- **X** is a set of variables, $\{X_1, \dots, X_n\}$. [**finite set of variables**]
- **D** is a set of domains, $\{D_1, \dots, D_n\}$, one for each variable.
- **C** A finite set of **constraints** ($C = \{C_1, C_2, C_3, \dots, C_n\}$)

Defining Constraint Satisfaction Problems

- Each constraint C_j consists of a pair $\langle \text{scope}, \text{relation} \rangle$
- **Scope** it is set of variables that participate in constraint
- **Relation** define values that variables can take.
- CSP is used intelligent backtracking algorithm to solve the problem.

Types of constraints

- There are many different types of constraints, and each is handled slightly differently:
- **Unary Constraints** - Unary constraints involve a single variable in the CSP.
- **Binary Constraints** - Binary constraints involve two variables.
- **Higher-order Constraints** Constraints involving three or more variables

Unary Constraint

Unary Constraint (involves a single variable)



Example: In a graph coloring problem, a unary constraint could be:

"Variable V1 (representing a node) cannot be assigned the color 'Red'."



Another example:

"Variable X (representing an exam time slot) must be scheduled before 12 PM."

2. Binary Constraint (involves two variables)



(graph coloring):

*"Adjacent nodes $V1$ and $V2$ cannot
have the same color."*

3. Higher-order Constraint (involves three or more variables)



Example (scheduling):

*"At least two out
of X, Y, Z (representing employees) must
be assigned the morning shift."*

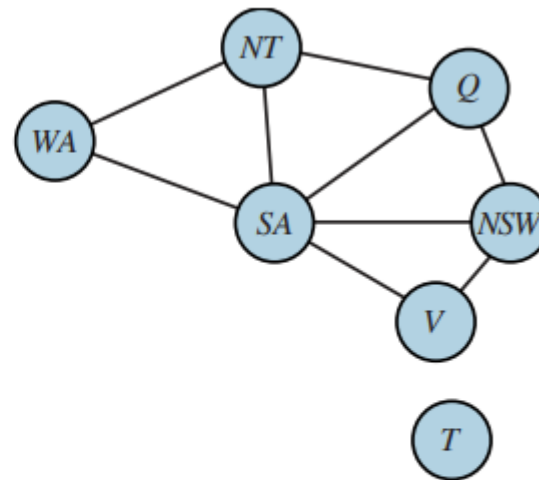
Example: Map-Coloring

- We are given the task of coloring each region of Australia either **red**, **green**, or **blue** in such a way that no neighboring regions have the same color.

Example: Map-Coloring



(a)



(b)

Figure 6.1 (a) The principal states and territories of Australia. Coloring this map can be viewed as a constraint satisfaction problem (CSP). The goal is to assign colors to each region so that no neighboring regions have the same color. (b) The map-coloring problem represented as a constraint graph.

Example: Map-Coloring

To formulation as a CSP,

Variables: Each region is a variable: $X = \{WA, NT, Q, NSW, V, SA, T\}$.

Domains: The domain of each variable is the set $D_i = \{\text{red}, \text{green}, \text{blue}\}$.

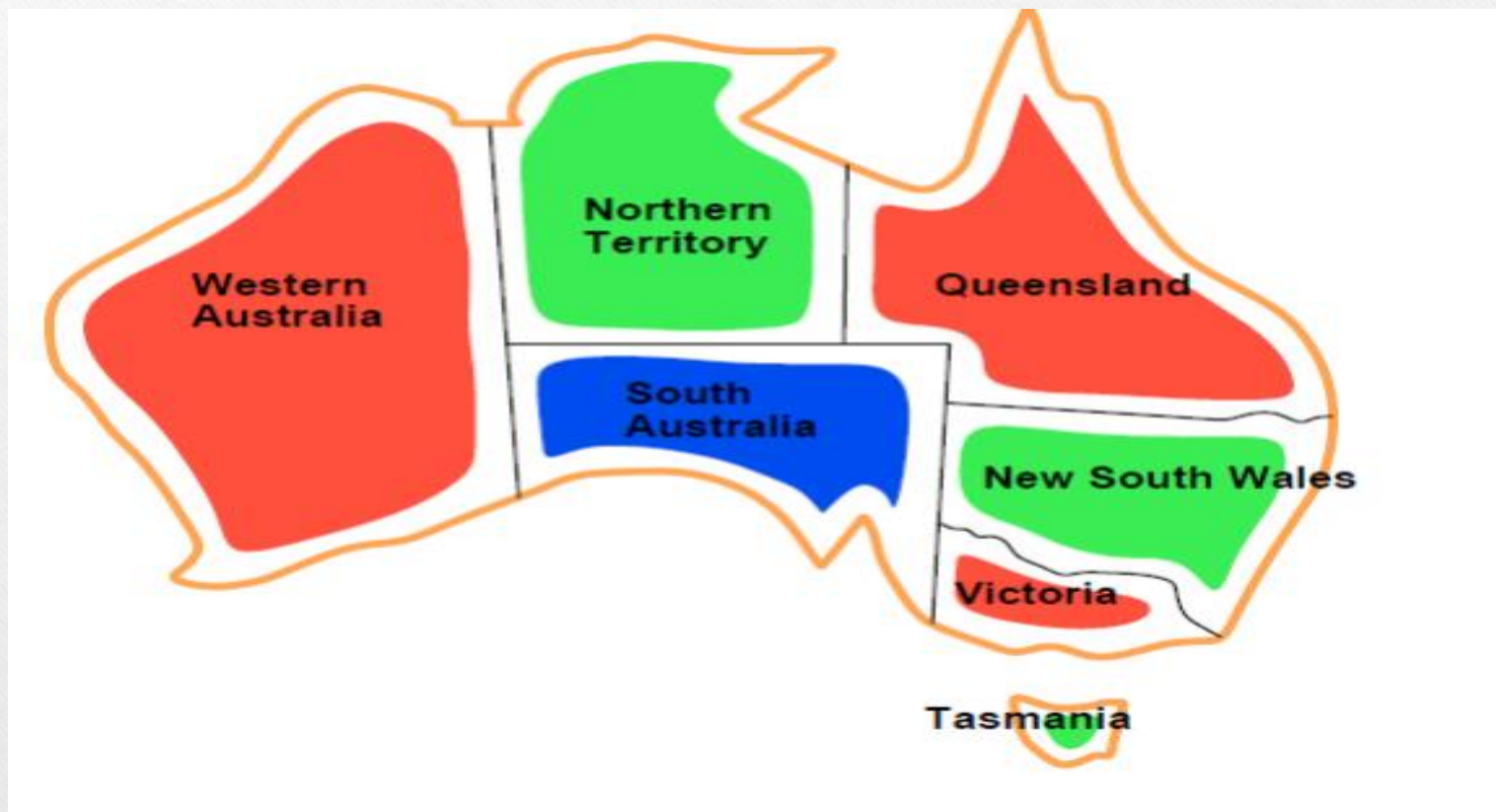
Example: Map-Coloring

- **Constraints** require neighboring regions to have distinct colors.
- Since there are nine places where regions border, there are nine constraints:
- $C = \{SA \neq WA, SA \neq NT, SA \neq Q, SA \neq NSW, SA \neq V, WA \neq NT, NT \neq Q, Q \neq NSW, NSW \neq V\} .$

Example: Map-Coloring

- Solutions are assignments satisfying all constraints, e.g., {WA=red, NT=green, Q=red, NSW=green, V=red, SA=blue, T=green}

Example: Map-Coloring



Real-world CSPs

- Assignment problems, e.g., who teaches what class?
- Timetabling problems, e.g., which class is offered when and where?
- Hardware configuration
- Spreadsheets
- Transportation scheduling
- Factory scheduling
- Floor planning

Variations on the CSP formalism

- The simplest kind of CSP involves variables that have **discrete, finite domains**.
- Map-coloring Discrete domain problems and scheduling with time limits are both of this kind.
- The 8-queens problem can also be viewed as a **finite-domain** CSP, where the variables $Q_1, \dots, Q_8$ correspond to the queens in columns 1 to 8, and the domain of each variable specifies the possible row numbers for the queen in that column, $D_i = \{1, 2, 3, 4, 5, 6, 7, 8\}$.
- The constraints say that no two queens can be in the same row or diagonal.

Two of our GSIs, Arjun and Woody, are making their schedules for a busy morning. There are five tasks to be carried out:

- (F) Pick up food for the group's research seminar, which, sadly, takes one precious hour.
- (H) Prepare homework questions, which takes 2 consecutive hours. (P) Prepare the PR2 robot for a group of preschoolers' visit, which takes one hour.
- (S) Lead the research seminar, which takes one hour.
- (T) Teach the preschoolers about the PR2 robot, which takes 2 consecutive hours.

The schedule consists of one-hour slots: 8am-9am, 9am-10am, 10am-11am, 11am-12pm. The requirements for the schedule are as follows:

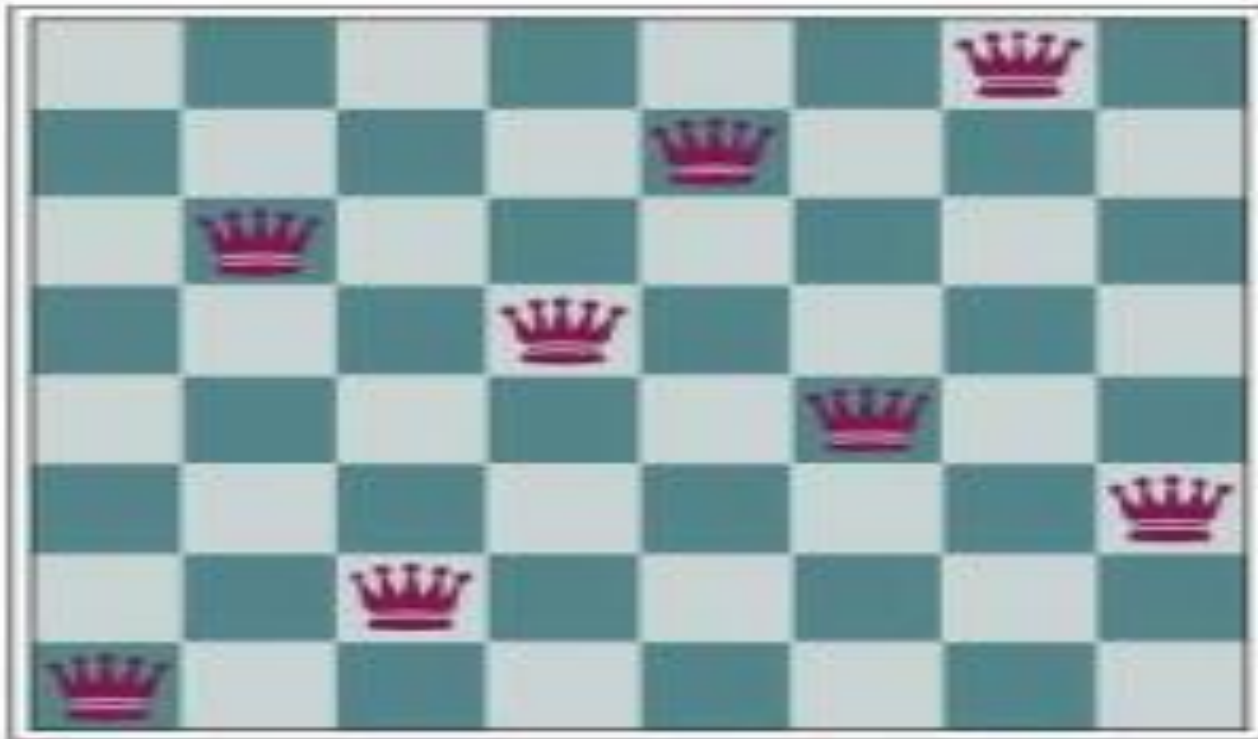
- (a) In any given time slot each GSI can do at most one task (F, H, P, S, T).
- (b) The PR2 preparation (P) should happen before teaching the preschoolers (T).
- (c) The food should be picked up (F) before the seminar (S).
- (d) The seminar (S) should be finished by 10am.
- (e) Arjun is going to deal with food pick up (F) since he has a car.
- (f) The GSI not leading the seminar (S) should still attend, and hence cannot perform another task (F, T, P, H) during the seminar.
- (g) The seminar (S) leader does not teach the preschoolers (T).
- (h) The GSI who teaches the preschoolers (T) must also prepare the PR2 robot (P).
- (i) Preparing homework questions (H) takes 2 consecutive hours, and hence should start at or before 10am.
- (j) Teaching the preschoolers (T) takes 2 consecutive hours, and hence should start at or before 10am. To formalize this problem as a CSP, use the variables F, H, P, S and T.

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- Which of the statements above includes unary constraints?
 - A unary constraint constrains the domain of a single variable. (d), (e), (i), (j) are unary constraints. (i) and (j) express both unary constraints (on the time of the tasks) and binary constraints (the length of tasks excludes other assignments during their time).

Variations on the CSP formalism

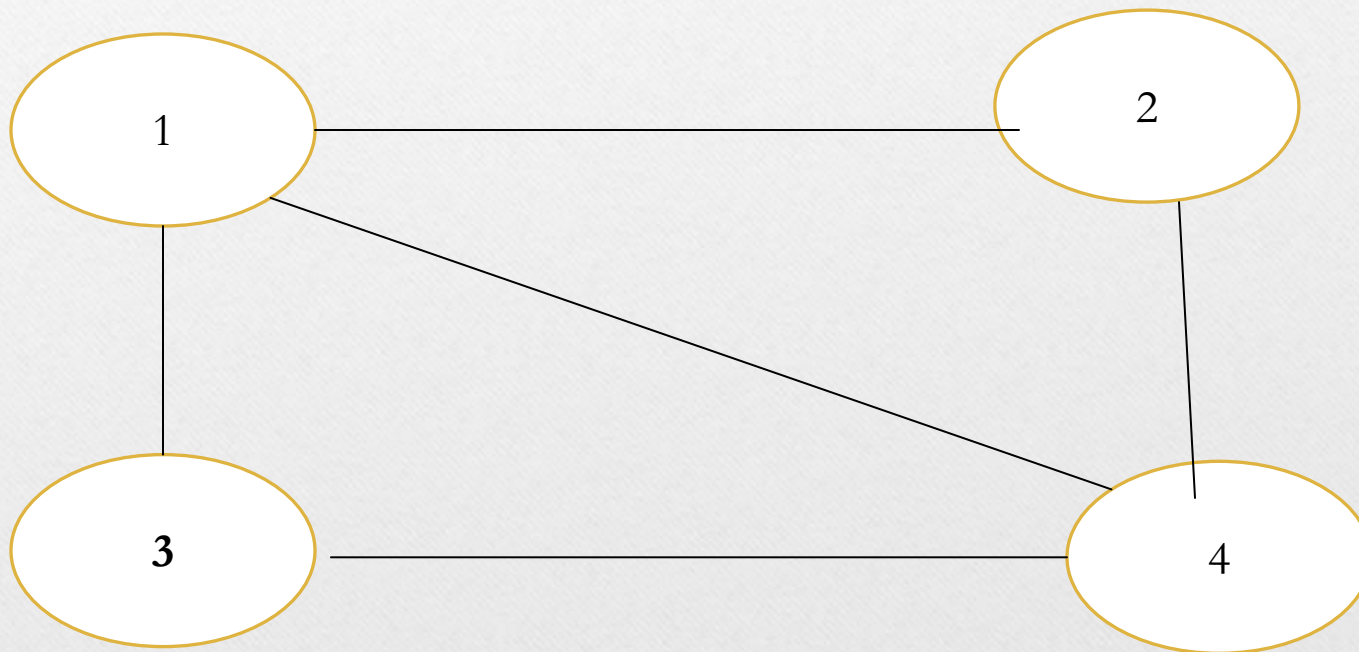
- A discrete domain can be infinite, such as the set of integers or strings. (If we didn't put Infinite a deadline on the job-scheduling problem, there would be an infinite number of start times for each variable.)
- A constraint involving an arbitrary number of variables is called a **global constraint**. E.g Cryptarithmic

The 8-queens problem:



(a)

Intelligent Backtracking in CSP



Intelligent Backtracking in CSP

- $V = \{1, 2, 3, 4\}$
- $D = \{\text{RED}, \text{GREEN}, \text{BLUE}\}$
- $C = \{\text{Adjacent nodes should not have same color}\}$

Intelligent Backtracking in CSP

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where

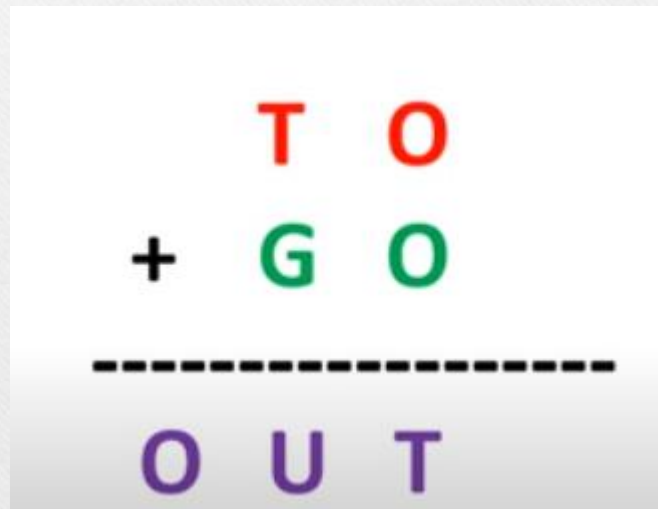
	1	2	3	4
Initial	R, G, B	R, G, B	R, G, B	R, G, B
1 = R	R	G, B	G, B	G, B
2 = G	R	G	G, B	B
3 = B	R	G	B	
3 = G	R	G	G	B

ERROR

Cryptarithmic problem Rules

- Digit ranges from 0 to 9 only.
- Each Variable should have unique and distinct value.
- Each Letter, Symbol represents only one digit throughout the problem.
- There must be only one solution to the problem.
- Numbers must not begin with zero i.e. 0123 (wrong), 123 (correct).

Cryptarithmic problem



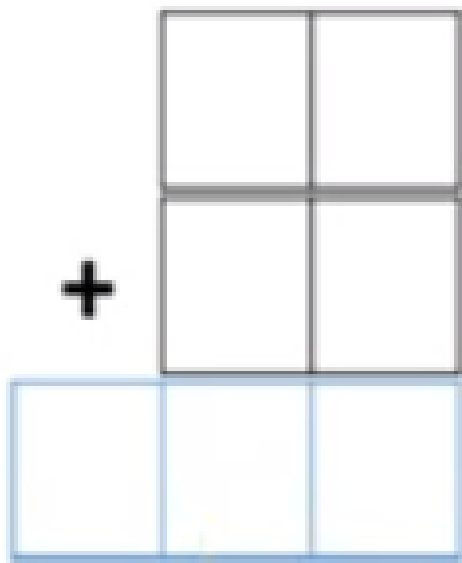
A cryptarithmic problem diagram showing the addition of two two-digit numbers to form a three-digit result. The first number is 'TO' in red, the second is 'GO' in green, and the result is 'OUT' in purple. A dashed line separates the addends from the sum. A plus sign is placed between the two addends. The entire diagram is centered on a light gray rectangular background.

$$\begin{array}{r} \text{TO} \\ + \text{GO} \\ \hline \text{OUT} \end{array}$$

Cryptarithmic problem

$$\begin{array}{r} \text{C1} \\ \text{T O} \\ + \text{G O} \\ \hline \text{O U T} \end{array}$$

Cryptarithmic problem



Character	Code
T	
O	
G	
U	

Cryptarithmic problem

$$\begin{array}{r} 1 \\ + 1 \\ \hline 1 \end{array}$$

O
O
T

Character	Code
T	
O	1
G	
U	

Cryptarithmic problem

$$\rightarrow 0 + 0 = T$$

$$\rightarrow 1 + 1 = T$$

$$\rightarrow T = 2$$

$$\begin{array}{r} 21 \\ + 1 \\ \hline 12 \end{array}$$

Character	Code
T	2
O	1
G	
U	

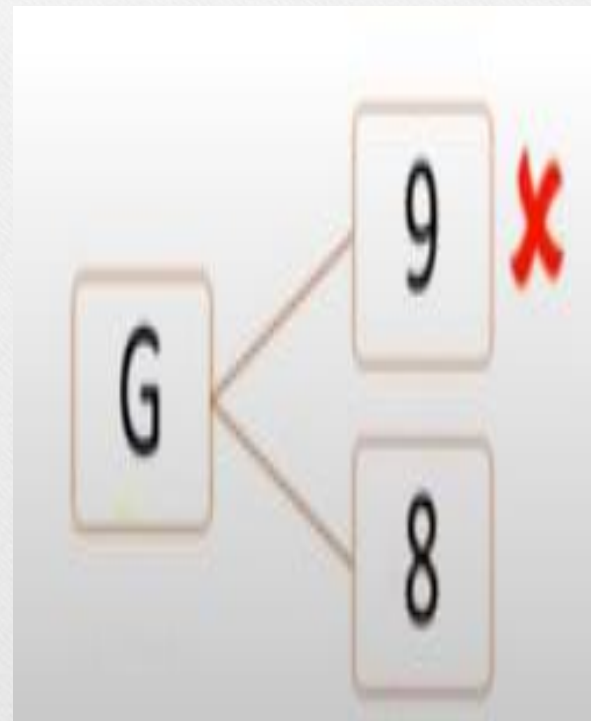
Cryptarithmic problem

$$\rightarrow T + G = U$$

$$\rightarrow 2 + G = U + 10$$

$$\rightarrow 2 + 9 = U + 10$$

$$\rightarrow U = 1 \text{ } \times$$



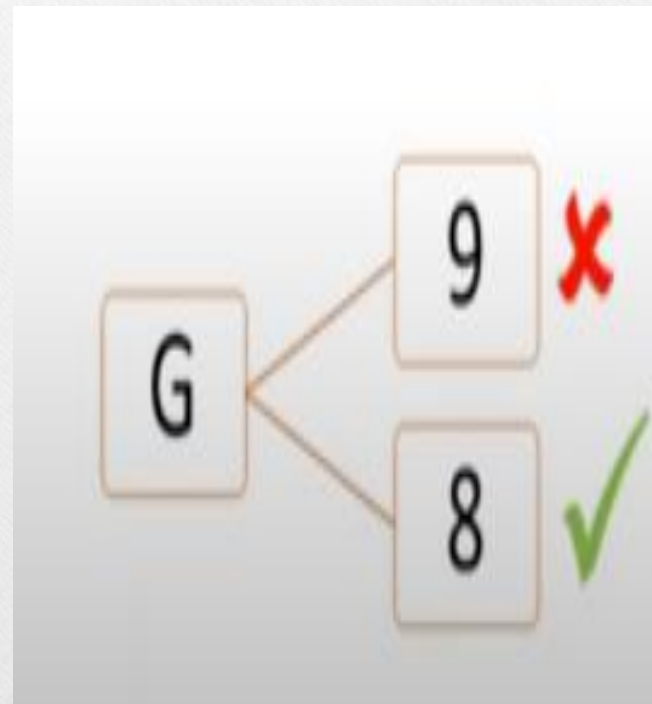
Cryptarithmic problem

$$\rightarrow T + G = U$$

$$\rightarrow 2 + G = U + 10$$

$$\rightarrow 2 + 8 = U + 10$$

$$\rightarrow U = 0$$



Cryptarithmic problem

C1

$$\begin{array}{r} \text{TO} \\ + \text{GO} \\ \hline \text{OUT} \end{array}$$

Character	Code
T	2
O	1
G	8
U	0

$$\begin{array}{r} 21 \\ + 81 \\ \hline 102 \end{array}$$